

Microcosm Sustainability Intelligence Report

2025 Data

Kumaraguru College of Technology

Generated on 14 July 2026

Compare Mode: YOY

Executive Summary

Performance Overview & Net Carbon Impact

During this reporting period, the campus generated 1753.80 tCO₂e of gross carbon emissions. Through our renewable energy initiatives, we successfully avoided 2452.44 tCO₂e, resulting in a net carbon impact of -698.64 tCO₂e. We remain committed to our long-term decarbonization pathway.

NET EMISSIONS

-698.64 tCO₂e

Final footprint after offsets

GROSS EMISSIONS

1753.80 tCO₂e

0.0% vs prev

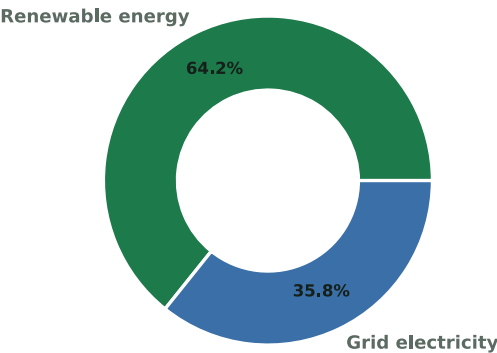
AVOIDED EMISSIONS

2452.44 tCO₂e

0.0% vs prev

Renewable Energy Integration

Renewable Mix



Strategic Observation

Renewable energy integration contributed 3373368.00 kWh, offsetting approximately 2452.44 tCO₂e of potential emissions. Our current renewable share stands at 64.2%.

RENEWABLE SHARE

64.2 %

Scope 1 Analysis

Direct Fuel Emissions from Operations

Scope 1 emissions generated 389.11 tCO₂e. Transport remains the dominant emission source in this category. We achieved a reduction of 0% compared to the previous period.

TOTAL SCOPE 1

389.11 tCO₂e

0.0% vs prev

DIESEL EMISSIONS

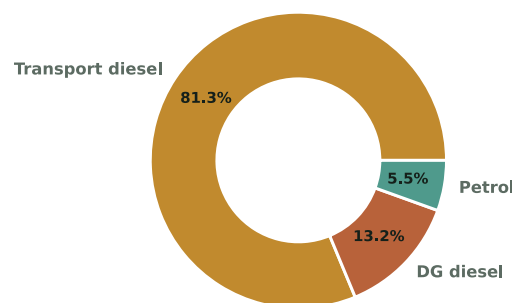
367.73 tCO₂e

PETROL EMISSIONS

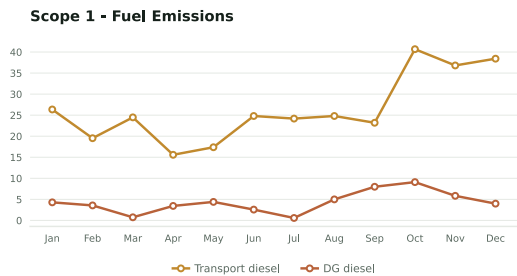
21.38 tCO₂e

Fuel Source Breakdown

Scope 1 Breakdown



Monthly Fuel Trends



Scope 2 Analysis

Indirect Emissions from Purchased Electricity

Grid electricity consumption accounted for 1364.69 tCO₂e. This represents 77.8% of total gross emissions. Continuous efforts in optimizing HVAC and lighting can further reduce this footprint.

TOTAL SCOPE 2

1364.69 tCO₂e

0.0% vs prev

TOTAL CONSUMPTION

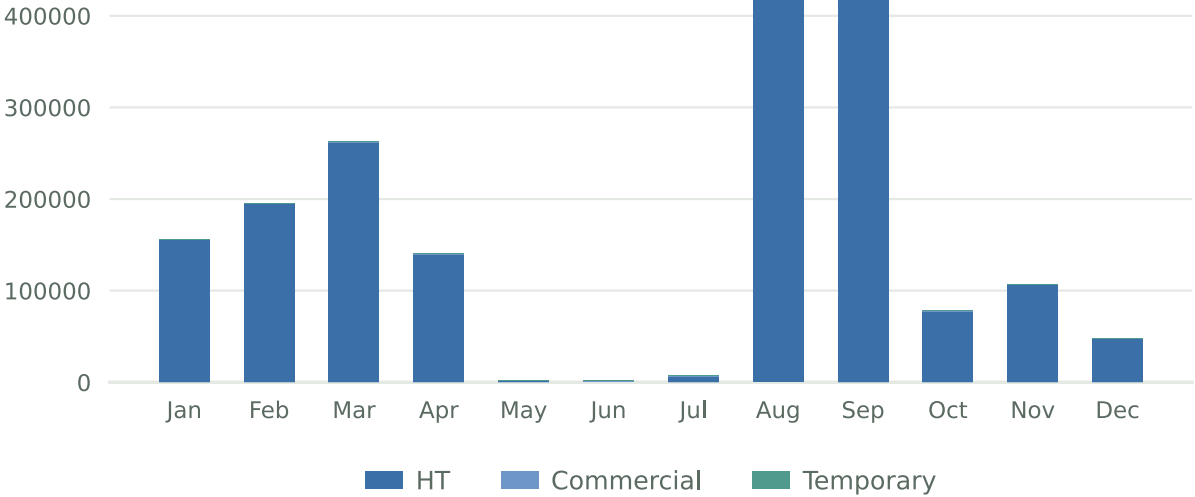
1877147 kWh

GRID CARBON FACTOR

0.727 kg/kWh

Electricity Consumption by Category

Scope 2 - Grid Consumption



Strategic Recommendations

Actionable Insights for Decarbonization

Based on the data collected and analyzed for the 2025 Data reporting period, the following recommendations are provided to accelerate our progress towards our net-zero targets.

Initiative 1

Accelerate the transition of internal campus mobility to Electric Vehicles (EVs) to reduce Scope 1 transport emissions.

Initiative 2

Expand rooftop solar capacity to push the renewable energy mix beyond the current baseline.

Initiative 3

Implement automated BMS (Building Management Systems) for high-load commercial blocks to optimize HVAC consumption.

Methodology & Appendix

Reporting Standards and Emission Factors

This Sustainability Intelligence Report has been generated automatically using verified utility and transport data. Emissions are calculated strictly utilizing recognized standard emission factors.

Standard Emission Factors Applied

EMISSION SOURCE	FACTOR	UNIT
Grid Electricity (TN)	0.727	kgCO ₂ e / kWh
Diesel Fuel	2.701	kgCO ₂ e / L
Petrol Fuel	2.388	kgCO ₂ e / L

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